

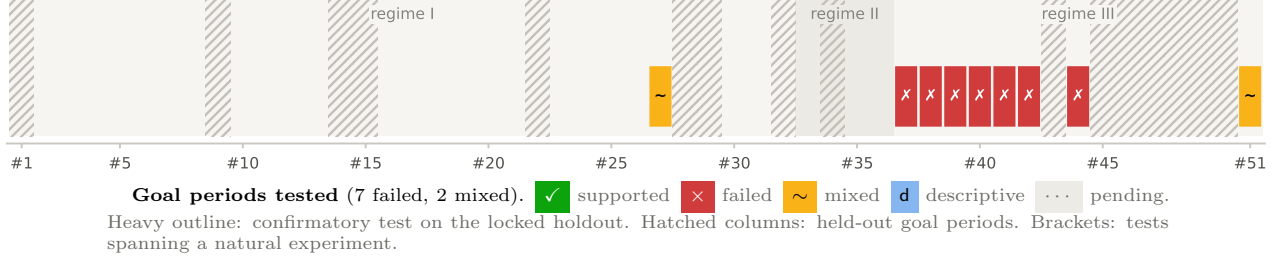
H14: Entropy production of behavior-state sequences: per-family arrows of time, and collective irreversibility

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-03

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Bottom line. Agent behavior is irreversible mainly at the level of fine tool actions and where the scaffold imposes order; per-family arrows (HH19) and collective irreversibility (HH67) were not found.

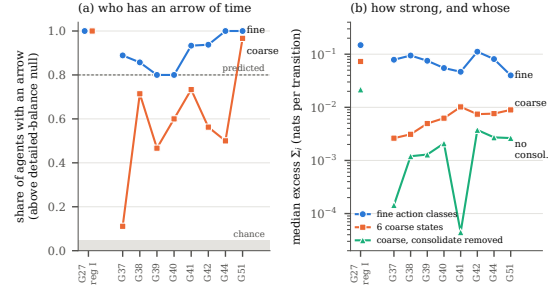
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WHAT HAPPENED

Is each agent's behavior statistically irreversible? Does each model family have its own arrow of time (HH19)? Is the swarm more irreversible than the sum of its agents (HH67)? Round 1 (exploratory, non-holdout) covered 9 goal periods: #27 (regime I), and #37–#44 and #51 (regime III). It built categorical states from computer-use turns and agent events. On 6 coarse states (browse, type, shell, chat, idle, consolidate), regime-III arrows are real but small: median 0.002–0.010 nats/transition. At least 80% of agents beat a detailed-balance null only in #51 (29/30) and #27 (10/10, with arrows 7–29× larger). On 11 fine action classes, 80–100% of agents are irreversible in every period. Removing the scaffold's consolidation step removes 52–100% of the coarse excess. There is no family effect (stratified $p = 0.89$), only a weak per-agent trait ($\rho = 0.13$ – 0.24 , $p \leq 0.02$), and no collective term after Holm. Verdicts: 7 failed, 2 mixed. Holdout not run.

$$\begin{aligned}\Sigma_i &= \sum_{a,b} \pi_a P_{ab} \ln \frac{\pi_a P_{ab}}{\pi_b P_{ba}}, & g_{ab} &= \mathbf{1}[a \rightarrow b] - \mathbf{1}[b \rightarrow a] \\ \Sigma_g &= \max_{\theta} [\theta \cdot \langle g \rangle - \ln \langle e^{-\theta \cdot g} \rangle], & \hat{\Sigma}_N &= 2 \langle \bar{g}_a^\top K^{-1} \bar{g}_b \rangle_{a \neq b} \\ P(x'_i = a | \mathbf{x}) &\propto e^{h_i^a + \sum_j J_{ij}^{a,x_j}}, & \Delta \Sigma &= \Sigma_{1+2} - \Sigma_1 \\ A_{abc} &= \ln \frac{P_{ab} P_{bc} P_{ca}}{P_{ac} P_{cb} P_{ba}}\end{aligned}$$



Observables. (a) Share of agents whose state sequence beats a detailed-balance surrogate (same occupancies and symmetric transition counts, currents removed). Almost all agents do on fine action classes; on the 6 coarse states only a minority do, except in the long #51 window and in regime I (#27). (b) Median excess irreversibility. Fine micro-workflow carries about 10× the coarse arrow, and most of the coarse arrow vanishes when the scaffold's consolidation step is removed.

PHYSICAL MODEL

Kinetic Potts (models 02, 10). $x_i(t) \in \{1..q\}$ is agent i 's behavior state per turn (single agent) or per 1-min bin (swarm). P_{ab} and π_a are its transition matrix and occupancies. Σ_i is the entropy production per transition: the pair KL, a lower bound for non-Markov agents. It is estimated with the Aguilera–Itō–Kolchinsky dual on antisymmetrized transition indicators g_{ab} , cross-fitted by day (Newton step $\hat{\Sigma}_N$). J_{ij}^{ab} are cross-agent couplings, $\Delta \Sigma$ is the collective increase of the bound, and A_{abc} a cycle affinity.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Mostly a careful measurement of the scaffold. The strongest arrows of time are GUI and tool micro-workflow (click, type, key), which is trivially irreversible. The coarse arrow is set by when the harness forces consolidation, or by regime I's session structure. Families don't differ, and there is no swarm-level irreversibility at a detectable size, so "irreversibility as a superagent signature" found nothing. What is practical: a calibrated detailed-balance null (shuffle nulls overstate irreversibility); a per-agent irreversibility score weakly stable across weeks (a fingerprint candidate); and a warning that cross-agent EP comparisons need null-subtracted estimators. An operator would change nothing yet. The Jev semantic states are the remaining chance for an agent-chosen arrow.

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